

Three example programs are provided in Part5/programs/:

- test_conditionals.cl
- test_loops.cl
- test_functions.cl

These examples collectively test all major language constructs.

Lexer & Parser Test Coverage Table

Example file	Line(s)	Tested phase	What is tested
test_conditionals.cl	1–5	Lexer + Parser	WORLD block and declarations: SIZE, ENTRY_DEF, EXIT_DEF, directions N, S, punctuation { } () , ;
test_conditionals.cl	7–9	Lexer + Parser	AGENT block header + variable declarations (VAR, identifiers, integer literals, =)
test_conditionals.cl	12–16	Lexer + Parser	IF / ELSE / ENDIF structure + comparison operator GT + REPORT
test_conditionals.cl	18–22	Lexer + Parser	Boolean expression with EQ, LT, and AND inside an IF
test_conditionals.cl	24–28	Lexer + Parser	Boolean expression with NEQ, EQ, and OR inside an IF
test_loops.cl	1–5	Lexer + Parser	WORLD block and declarations
test_loops.cl	7–9	Lexer + Parser	AGENT header + variable declarations (counter, max)
test_loops.cl	11–14	Lexer + Parser	WHILE ... DO ... ENDWHILE loop + relational operator LT
test_loops.cl	12–13	Lexer + Parser	REPORT with identifier + assignment statement and arithmetic +
test_functions.cl	1–5	Lexer + Parser	WORLD block and declarations
test_functions.cl	7–9	Lexer + Parser	Function definition syntax: FUNC, parameters, RETURNS INT, RETURN, expression A + B
test_functions.cl	11–14	Lexer + Parser	AGENT header + variable declarations (result1, result2)
test_functions.cl	15–16	Lexer + Parser	Function call in assignment: result1 = add(2, 3); then REPORT result1;
test_functions.cl	18–19	Lexer + Parser	Function call in assignment: result2 = add(10, 5); then REPORT result2;
test_functions.cl	21	Lexer + Parser	Nested function call in REPORT add(result1, result2);